MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 12000012

(COMPUTATION CONTROL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Computation Control Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-50785 | Circuit Card Assembly, Sight Current Generator, General Specification for |

DRAWINGS

U.S. Army, Frankford Arsenal

| | |
|---|---|
| 12000239 | Sight Current Generator |
| 10545455 | Receptacle |
| 12000012 | Circuit Card Assembly, Computation Control |

PACKAGING DATA SHEET

| | |
|---|---|
| 12000012 | Packaging of Circuit Card Assembly, Computation Control |

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Item definition.- This assembly is a single-ended printed circuit board consisting of a triangle wave generator and two inverters. The assembly is used to invert and/or amplify voltages within the Sight Current Generator Drawing Number 12000239 in which it is assembled.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000012.

3.1.2 Government furnished property list. Not applicable.

3.2 Characteristics.

3.2.1 Performance.- The assembly shall be capable of performing functions a through c when assembled in accordance with Drawing Number 12000012.

a. Inverter (Q1) shall invert one of the inputs required to control the "Ready to Fire" circuitry and "Acquisition Time Delay" circuitry.

b. Inverter (Q2) shall invert one of the inputs required to control the "Ready to Fire" circuitry.

c. The triangle wave generator (AR1 - AR2) shall generate an 18 volt P-P triangular wave within a period of approximately 0.8 millisecond.

3.2.1.1 Input and output levels.

3.2.1.1.1 Input power.- Input power shall be supplied to the assembly as specified in Table I.

TABLE I - INPUT VOLTAGES

| Pin | Voltage |
|---|---|
| 4 and 25 | +15 Vdc ± 2.0 percent |
| 14 | -15 Vdc ± 2.0 percent |
| 1 and 28 | Ground |

3.2.1.1.2 Output levels.- The requirements specified in 3.2.1.1.2.1 through 3.2.1.1.2.3 shall be met when supplied with the specified voltage input.

3.2.1.1.2.1 Pin 26 output.- The output shall be +15 Vdc ± 2.0 percent between pin 26 and pin 28 with zero Vdc input applied between pin 18 and pin 28, and 0.5 ± 0.5 Vdc with +15 Vdc input applied between pin 18 and pin 28.



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3.2.1.1.2.2 Pin 27 output.- The output between pin 27 and pin 28 shall be as specified in a through c.

a. As shown in Figure 1 when +15 Vdc is removed from pin 17.

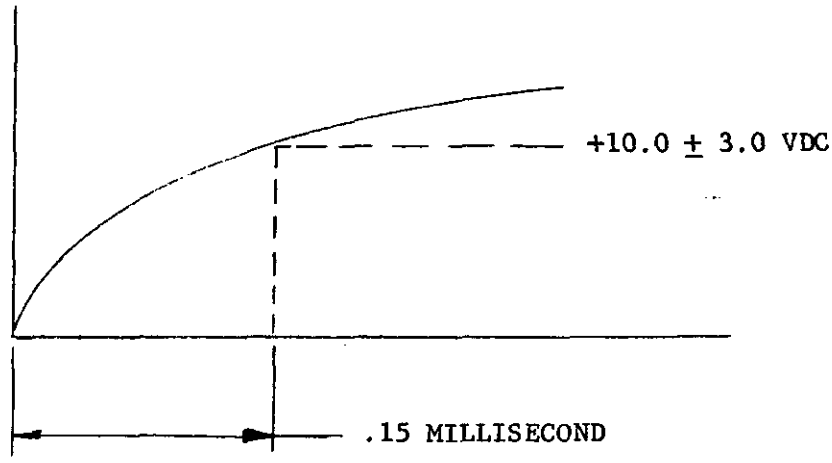



Figure 1.

b. +15 Vdc ± 4.0 percent with zero Vdc applied to pin 17.

c. +1.0 ± 0.5 Vdc with +15 Vdc applied between pin 17 and pin 28.

3.2.1.1.2.3 Pin 3 output.- The output shall be a triangular wave with an amplitude from -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of 0.8 ± 0.11 milliseconds.

3.2.1.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristics.- The physical characteristics shall be as specified in drawing 12000012 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 12000012 and associated lists.

3.3.2 Standards of manufacture.- The standards of manufacture shall be those specified in specification MIL-C-50785.

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3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-50785.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment.

4.2.1.1 Power supply (PS1).- The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS2).- The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.3 Switches.- Two single pole single throw switches are required.

4.2.1.4 Oscilloscope.- The oscilloscope shall have voltages, time accuracy and resolution of 5.0 percent or better, with a rise time of 100 nanoseconds or better and with external triggering. Voltage sensitivity shall include the range from 0.05 volt/cm to 5.0 volts/cm.

4.2.1.5 Digital voltmeter (DVM).- The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

4.2.1.6 Resistor.- One resistor is required with a rating of 1100 ohms ± 5.0 percent, 2.0 watts.

4.2.2 Functional test.

4.2.2.1 Test connections.- Power supplies PS1 and PS2 shall be connected to the assembly under test in accordance with Figure 2 using receptacle 10545455.



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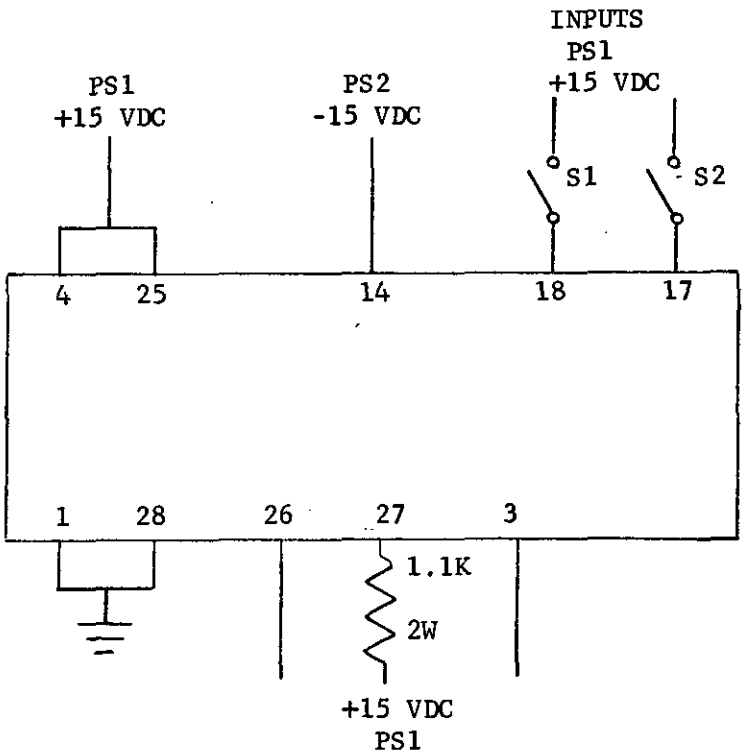



Figure 2. Test Connections

4.2.2.2 Test procedure.- Compliance with the requirements of 3.2.1.1.2 through 3.2.1.1.2.3 shall be verified as follows:

a. With switch S1 off, the voltage at pin 26 measured with the DVM shall be +15 Vdc ± 2.0 percent.

b. With switch S1 on, the voltage at pin 26 measured with the DVM shall be 0.5 ± 0.5 Vdc.

c. With switch S2 off, the voltage at pin 27 measured with the DVM shall be +15 Vdc ± 4.0 percent.

d. With switch S2 on, the voltage at pin 27 measured with the DVM shall be +1.0 ± 0.5 Vdc.

e. When the switch S2 is positioned from on to off, the voltage at pin 27 measured with the oscilloscope shall be as shown in Figure 1.

f. With the oscilloscope connected to pin 3, the voltage shall be a triangular wave with an amplitude from -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of 0.8 ± 0.11 milliseconds.



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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 12000012.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A431



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-C-50785/12(MU) Circuit Card Assembly: 12000012 (Computation Control)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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